GY462—Real Estate Finance

Lecture 7

1

Effective Cost of Borrowing

- **Additional costs**
  - **Origination fees in form of 'discount points' (or 'prepaid interest')**
  - **Other origination charges**
  - **Prepayment penalties**

⇨ **Drive wedge between YTM and contractual rate**

2

Discount Points and Effective Cost —*Example*

- **Consider CPM of 1,000,000**
  - **8% annual interest**
  - **Repaid monthly in 30 years**
  - **1 discount point**
- **YTM (IRR) if no discount points?**
  - **Identical to contractual rate of** *8%/12 = 0.667%*
- **Effective cost of borrowing?**

3

With the Calculator…

Key: Use Time Value of Money Worksheet

2nd [RESET] ENTER (Sets all variables to defaults)

2nd [P/Y] 12 ENTER (Defines monthly compounding)

CE/C (Exit defining compounding period)

30 2nd [xP/Y] N (Enter number of payments)

8 I/Y (Enter annual interest rate)

1000000 +/- PV (Enter loan amount)

CPT PMT (Computes payment)

Display: *PMT* = 7,337.65 *

4

© Olmo Silva
Reading these notes without attending the relevant lectures can be misleading.

With the Calculator…

Key: Select Cash Flow Worksheet Data

[2nd] [FORMAT] 4 [ENTER] [CE/C] (Show 4 decimals)

[CF] (Select Cash Flow Worksheet)

990,000 [+/-] [ENTER] (Enter initial CF)

[↓] 7337.65 [ENTER]

[↓] 360 [ENTER] (Enter CFs for months 1-360)

[IRR] [CPT] (Compute IRR (~Yield to Maturity))

Display: $IRR = 0.6755\%$ * $\Rightarrow$ YTM (0.6755%) > i (0.6667%)

Early Repayment

- **Now suppose you can repay loan after 5 years**
  - **Compute remaining balance: 950,698.25**

$BAL=7,337.65\times(1/(1+0.08/12)+\ldots+1/(1+0.08/12)^{300})$

  - **Compute IRR: 8.25% per annum**

$990,000=7,337.65\times(1/(1+IRR)+\ldots+1/(1+IRR)^{60})$
$+\ 950,698.25/(1+IRR)^{60}$

With the Calculator…

[2nd] [RESET] [ENTER] (Sets all variables to defaults)

[2nd] [P/Y] 12[ENTER] [CE/C] (Defines monthly compounding)

30 [2nd] [xP/Y] [N] (Enter number of payments)

8 [I/Y] (Enter annual interest rate)

1000000 [+/-] [PV] (Enter loan amount)

[CPT] [PMT] (Computes payment) Display: $PMT = 7337.65$ *

[2nd] [AMORT]

[↓] 60 [ENTER] (Set ending period to 60)

[↓] Display: $BAL = -950,698.25$ *

With the Calculator…

[2nd] [RESET] [ENTER] (Sets all variables to defaults)

[2nd] [FORMAT] 4 [ENTER] (Show 4 decimals)

[CF] (Select Cash Flow Worksheet)

990,000 [+/-] [ENTER] (Enter initial CF)

[↓] 7337.65 [ENTER]

[↓] 59 [ENTER] (Enter CFs for months 1-59)

[↓] 950698.25 [+] 7337.65 [=] [ENTER] (Enter CF for month 60)

[IRR] [CPT] (Compute IRR (~Yield to Maturity))

Display: $IRR = 0.6875\%$ * $\Rightarrow$ $IRR\ p.a.$ = 8.25%

© Olmo Silva
Reading these notes without attending the relevant lectures can be misleading.

'Focus point 1'

Effective cost with prepayment: alternative solution?

9

Contents — *Lecture 7…*

3. Fixed Rate Mortgage Loans *(Cont.)*
   - Effective Cost of Borrowing
     - Discount Points
     - Prepayment Penalties
     - (Yield Maintenance)
4. **Mortgage Payment Issues**
   - **Inflation**
   - Reducing Monthly Payments *(GPMs, ARMs)*

10

Example: PMT and Inflation

- **£60,000 CPM 30 year mortgage at 4% (no inflation)**
  - **PMT = 286.45**
  - **Real value of PMT: Constant**
- **Instead: Expect 6% inflation**
  - ⇨ **Mortgage rate: (1.04×1.06) – 1 = 10.24%**
  - ⇨ **PMT = 537.21** *(⇨ +87.54%)*

11

Example: PMT and Inflation

- **Higher inflation ⇨ Higher nominal rate**

  ⇨ **Higher nominal PMT**

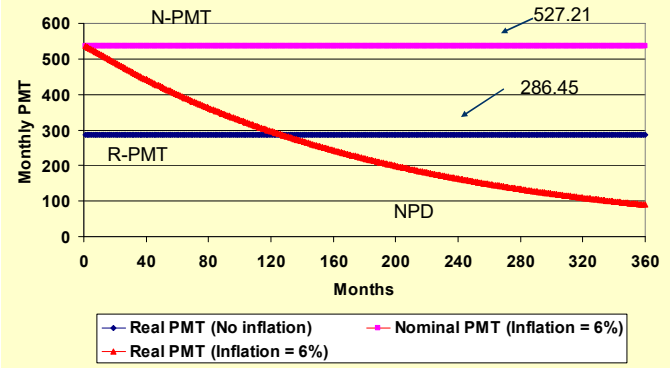
  ⇨ **PMT toward maturity worth less**
  - **Nominal value progressively eroded by inflation**

  ⇨ **<u>Real PMT higher early on</u>**

12

© Olmo Silva
Reading these notes without attending the relevant lectures can be misleading.

Payment and Inflation



Inflation's Impact on Borrower

- **Nominal and real PMT higher initially**
  - **Less households qualify**
- **Effect of inflation is non-linear**
  - **Small rate of inflation causes small 'tilt problem'**
  - **Large rate of inflation causes extremely large tilt problem**

How Can 'Tilt Problem' be Reduced?

⇨**Look at alternative mortgage types...**

To Reduce Monthly Payments

- **Three possible alternatives**
  - **Reduced initial monthly payments**
    - **Graduated Payment Mortgages (GPM)**
  - **Share interest rate risk**
    - **Adjustable Rate Mortgages (ARM)**
    - **(Shared Appreciation Mortgages – won't discuss; can look at B&F book)**

© Olmo Silva
Reading these notes without attending the relevant lectures can be misleading.

Graduated Payment Mortgage (GPM)

- PMT rises gradually then levels off
- PMT below CPM payment initially

⇨ Eventually, PMT exceeds CPM payment in order to compensate lower balance decline in early years

Negative Amortization on GPM – *A Simple Example*

- 60,000 GPM 30 year mortgage at 12% GPM payments
  - PMT should be 617.17 but
  - You are asked to pay 474.83 in first year
- AMORT first month
  - Interest: 60,000 × 0.12/12 = 600
  - AMORT = – 600 + 474.83 = –125.17

Negative Amortization on GPM – *Continued*

- Loan balance at end of first year?
  - Compound effect of negative amortization
  - Still charged at 1% a month
  - As if: 12 month loan of 60,000 at 12% with PMT of 474.83

    ⇒ BAL = 61,587

With the Calculator…

[2nd] [RESET] [ENTER] (Sets all variables to defaults)

[2nd] [P/Y] 12 [ENTER] [CE/C] (Monthly compounding)

12 [N] (Enter number of payments)

12 [I/Y] (Enter annual interest rate)

60000 [+/-] [PV] (Enter loan amount)

474.83 [PMT] (Enter payment)

[CPT] [FV] (Computes balance after one year)

Display: *FV* = 61,587.47 *

© Olmo Silva
Reading these notes without attending the relevant lectures can be misleading.

'Focus point 2'

GPM and negative amortization:
is the intuition clear?
And the calculations?

21

How to Determine Payments on GPM?

- Typical GPM
  - 7.5% annual graduation in monthly PMT over 5 years
  - Year 6 to 30 constant PMT

22

Payments on GPM – *Continued*

- Example
  Graduated 7.5% p.a., 5 years
  - 60,000 GPM
  - 30 year mortgage
  - Annual interest rate: 12%

Graduated Payment Mortgage	
PMT Year 1	474.83
PMT Year 2	510.44
PMT Year 3	548.72
PMT Year 4	589.87
PMT Year 5	634.11
PMT Year 6-30	681.67
CPM Equivalent PMT Years 1-30	617.17

23

Details of Payments on GPM

- How to determine first payment of GPM?
  - Think of mortgage as a cash flow
  - PV of cash flow is amount of money borrowed
  - Payments grow at predetermined stages
  - To determine initial payment, think of loan as a 'grouped' DCF
  - With initial PMT being unknown…

24

© Olmo Silva
Reading these notes without attending the relevant lectures can be misleading.

Details of Payments on GPM – *The Maths*

$$PV = \left[MP_1 \times \sum_{t=1}^{12} \frac{1}{(1+i/12)^t} \right] +$$ **First 12 months**

$$\left[MP_1 (1+g)^1 \times \sum_{t=1}^{12} \frac{1}{(1+i/12)^t} \frac{1}{(1+i/12)^{12}} \right] +$$ **13-24 months**

$$\left[MP_1 (1+g)^2 \times \sum_{t=1}^{12} \frac{1}{(1+i/12)^t} \frac{1}{(1+i/12)^{24}} \right] +$$ **25-36 months**

$$\left[MP_1 (1+g)^3 \times \sum_{t=1}^{12} \frac{1}{(1+i/12)^t} \frac{1}{(1+i/12)^{36}} \right] +$$ **37-48 months**

$$\left[MP_1 (1+g)^4 \times \sum_{t=1}^{12} \frac{1}{(1+i/12)^t} \frac{1}{(1+i/12)^{48}} \right] +$$ **49-60 months**

$$\left[MP_1 (1+g)^5 \times \sum_{t=1}^{300} \frac{1}{(1+i/12)^t} \frac{1}{(1+i/12)^{60}} \right] +$$ **61-360 months**

Details of Payments on GPM – *Details of the Maths*

- **How do you get discount factors and summations right?**
  - **$(1+g)^n$ term is easy…**
  - **Last component is not dependent on summation: 'take it out' of sum sign**
  - **Middle part is simply PV of an annuity paying 1 for 12 or 300 periods**
  - **PMT(1) equals PV of loan (i.e. mortgage) divided sum of all discount factors**
- **Important to understand mechanics; check carefully in B&F book**

'Focus point 3'

GPM as a 'grouped' DCF:
try the calculations – are they clear?

Check IRR of GPM with your Calculator…

… [RESET] … [FORMAT] 4 [P/Y] 12 …

[CF] (Select Cash Flow Worksheet)

-60000 [+/-] [ENTER] (Enter initial CF)

[↓] 474.83 [ENTER]

[↓] 12 [ENTER] (Enter CFs for months 1-12)

…

[↓] 681.67 [ENTER]

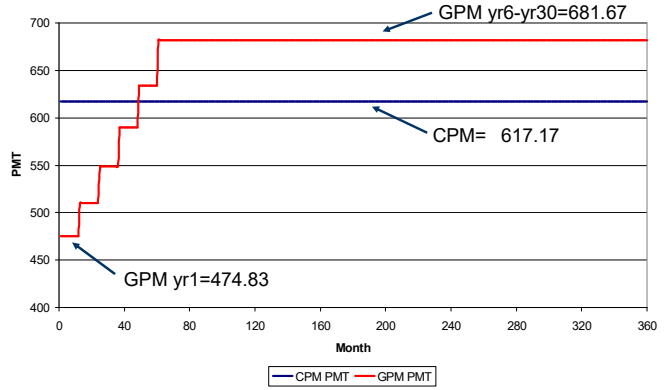
[↓] 300 [ENTER] (Enter CFs for months 61-360)

[IRR] [CPT] (Compute IRR)

Display: *IRR* = 1.0000% * ⇒ *IRR p.a.* = 12% q.e.d.

© Olmo Silva
Reading these notes without attending the relevant lectures can be misleading.

Payments: GPM versus CPM

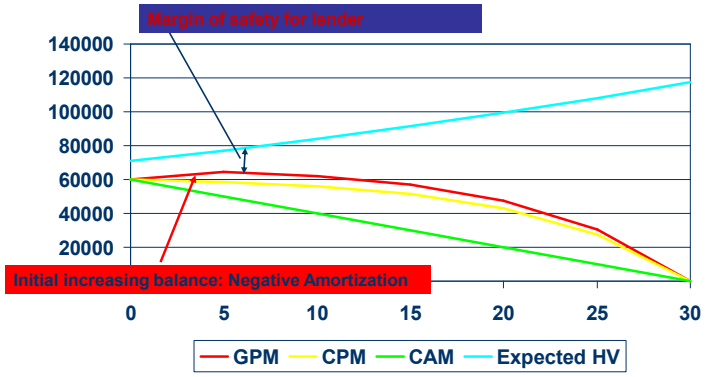


Lender's Point of View

- **Outstanding loan balance increases during initial years**
- **Makes GPM loan riskier**
  - **Smaller margin of safety**
  - **Uncertainty with regard to future HV and borrower income more relevant**

⇨ **GPM usually requires <u>higher risk premium</u> than CPM**

Loan Balance and House Value



Alternative mortgage types...

⇨ **Adjustable Rate Mortgages**

© Olmo Silva
Reading these notes without attending the relevant lectures can be misleading.

Adjustable Rate Mortgages (ARM) – *Why?*

- **Sharing interest rate risk between borrower and lender**
  - **Lender does not charge interest rate risk premium**
  - **Lower price for borrower**
  - **Why 'shared' risk?**

Adjustable Rate Mortgages (ARM) — *Key Terms*

- **ARMs typically described by**
  - **An index rate used as benchmark**
  - **Margin**
  - **Adjustment interval**
  - **Interest rate (or payment) caps**

ARM Payment Mechanics

- **Borrow 4 million, no payment caps, no interest rate caps, 2% margin, annual adjustment, 2 year maturity**
- **Initial index rate: 26%**
- **Year 1 payment** *(compounded annually)*:

As if CPM of 4 million over 2 years at 28%
⇨ PMT_1= 2.8744

Mortgage Balance on ARM

- **End of year balance**
  - **PV of remaining payments discounted at interest rate applicable during the year for which the balance is computed**
- **End of year 1 balance in example:**

As if CPM of 4 over 2 years at 28% after PMT of 2.8744
⇨ BAL_1= 2.2456

© Olmo Silva
Reading these notes without attending the relevant lectures can be misleading.

Subsequent Years

- New payments computed based on
  - New interest rate (index plus margin) at the end of adjustment interval
  - New loan balance
- Suppose year 2 index rate in example: 18%
- Year 2 payment:

  As if 2.2456 CPM 1 year loan at 20%

  ⇨ PMT_2= 2.6947

'Focus point 4'

ARM calculations: all clear?
And how would you approach these with monthly compounding?

Effective Cost of Borrowing: Unrestricted ARMs

- Borrow 4 million
- Payments: 2.8744 in year 1, 2.6947 in year 2
- Compute IRR: 25.53%

  $4 = 2.8744 /(1+IRR) + 2.6947/(1+IRR)^2$

- Interest Rate Risk Premium:

  = Yield on CPM – Expected Yield on ARM

Payment Mechanics – *With Interest Rate Caps*

- Interest rate caps limit change in ARM rate between adjustment periods
- Borrow $4 million for 2 years, 2% margin, 2% annual cap
- Initial index rate: 26%
- Year 1 payment:

  As if CPM of 4 over 2 years at 28%

  ⇨ PMT_1= 2.8744

© Olmo Silva
Reading these notes without attending the relevant lectures can be misleading.

Subsequent Years with Interest Rate Caps

- Index rate at start of year 2: 31%
- 2% annual cap ⇒ interest rate of 30%
- Year 2 payment:

  As if 2.2456 CPM 1 year loan at 30%

  ⇨ PMT_2= 2.9193

Payment Mechanics – *With Payment Caps*

- Payment caps limit increase in payments between adjustment periods
- Borrow $4 million for 2 years, 2% margin, 2% payment cap
- Initial index rate: 26%
- Payment in year 1:

  As if CPM of 4 over 2 years at 28%

  ⇨ PMT_1= 2.8744

Subsequent Years with Payment Caps

- Index rate at start of year 2: 38%
- Unrestricted Payment in Year 2:

  3.1439 = 2.2456 × 1.40
- Increase in payment:

  9.4% = (3.1439 / 2.8744)-1 *(>2%)*
- Thus, cap is binding, new payment is:

  2.8744 × 1.02 = 2.9319 (instead of 3.1439)

Problem?

- Amortisation is not sufficient to reduce balance to zero
  - AMT = Capped Payment – Interest
  - 2.0337 = 2.9319 – (2.2456 x 0.40)
- New loan balance at end of period 2 is 0.2120
  - BAL_2= BAL_1 – AMT = 2.2456 – 2.0337 = 0.2119

⇨ Borrower and lender must agree to extent life of loan

© Olmo Silva
Reading these notes without attending the relevant lectures can be misleading.

What next – *not much left…*

- ⇨ **PE / Development Case Study**
- ⇨ **Mortgage-Backed Securities**
- ⇨ **ESG in real estate investment**

- ❑ **Case study presentations in Week 11**
  - ✓ **Materials received?**
  - ✓ **Groups working well?**

London School of Economics—© Olmo Silva 45

45